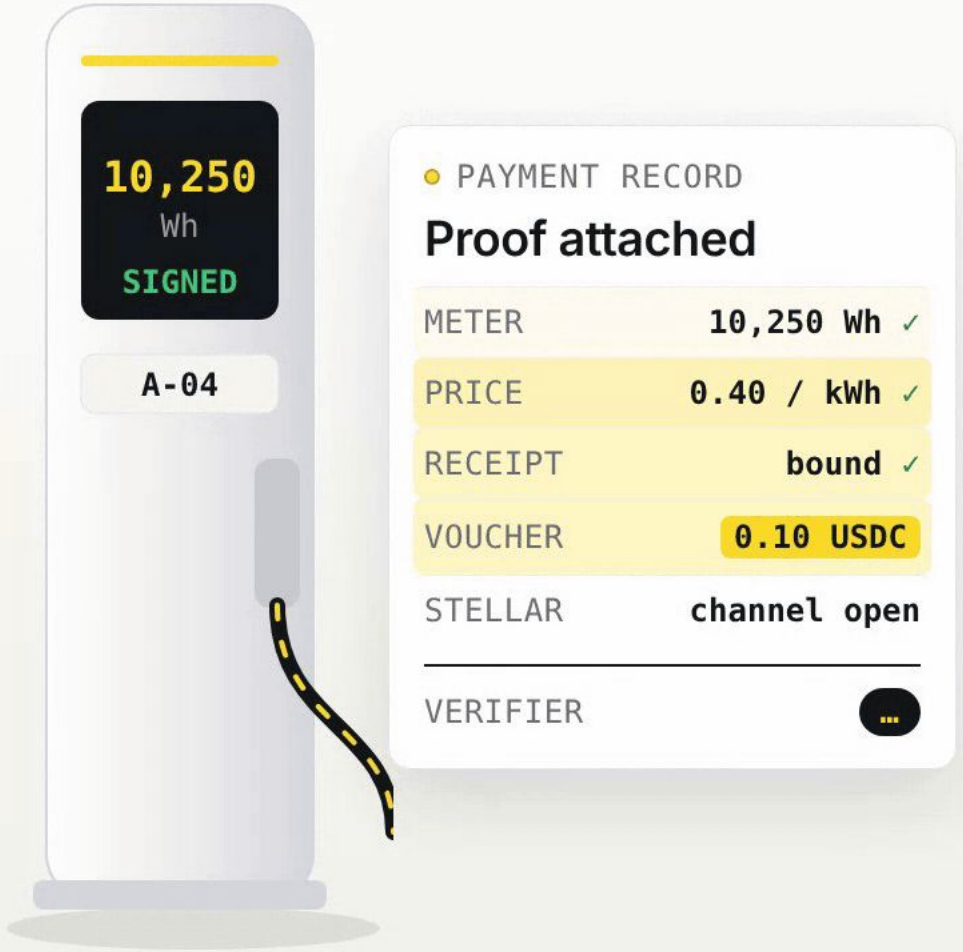
UsagePay on Stellar

Infrastructure for the machine economy

UsagePay enables verifiable machine-to-machine payments for metered services on Stellar.

Live on Stellar Testnet | How it works ↓

USE CASE: EV CHARGING · BUILT FOR ANY METERED SERVICE



Fleets pay for charging they can’t verify.

THE CHARGER SAYS

60 kWh

Recorded on the station’s system.

THE FLEET PAID

$25.20

Recorded on the fleet’s system.

PROOF THEY MATCH

None

Nothing links the two.

A payment proves **how much** and **to whom** — never **what for**. It’s a bank statement without the receipt.

UsagePay staples a receipt to each payment.

1

The charger signs

Every kWh it delivers is signed at the source.

2

The fleet pays

Only for signed kWh, at the agreed price.

3

Anyone checks

Public proof. No keys, no trust needed.

Someone changes a number? The signature breaks — and the payment never happens.

Runs on Stellar payment channels. No new hardware: chargers already sign their readings.

One fleet. Millions paid for charging.

EXAMPLE · AMAZON'S ELECTRIC DELIVERY FLEET

50,000+	electric delivery vans on the road
1 billion+	miles driven by its Rivian vans alone
$80–170M	paid for that charging (≈ 400M kWh, our estimate)
$1.7M	**at risk if just 1% can't be proven.** For one company.

70%
of fleet charging money goes to chargers the fleet doesn't own (UK).

100,000
Amazon e-vans planned by 2030. And that's one company.

UsagePay: every kWh paid is a kWh proven.

SOURCES: SUPPLYCHAIN247 (AUG 2026) · RIGHTCHARGE (H1 2026) · AAA (OCT 2026) · ESTIMATES ARE OURS

It already works — on Stellar.

3 → 1

Three charges paid with one single Stellar transaction.

Blocked

A reading changed by one digit never gets paid.

8/8

Independent checks passed. No keys needed.

Try it yourself: usagepay.vercel.app

Real run on Stellar Testnet, 25 Oct 2026 · transaction 0626430c…5ca1da ↗ · simulated meter · test USDC

UsagePay on Stellar

Pay only for what's proven.

UsagePay staples the receipt to every machine payment.

TRY THE LIVE DEMO

usagepay.vercel.app

NEXT

A pilot with one fleet and one charging network.

Open source: github.com/skopolabs/usagepay

WESLEY SANTOS · MARIO GARCÍA

Off-chain: the proof. On-chain: the money.

OFF-CHAIN · SIGNED MESSAGES · NO FEES

ON-CHAIN · STELLAR

1

The meter

Signs every reading with its own key.

2

The fleet

Checks signature, price and budget — then signs the MPP voucher and a receipt.

3

The station

Checks the receipt before accepting. The evidence is sealed (E0).

4

Stellar

One channel close pays the latest voucher. E1 links it to E0.

5 **Anyone can verify — no keys needed**

It re-checks every signature and price, and reads the chain itself through Stellar RPC.

STELLAR COMPOSED · REUSED UNCHANGED: MPP SDK · ONE-WAY CHANNEL · STELLAR RPC · NEW: THE RECEIPT + THE VERIFIER

Do not trust us: ask the ledger.

THE REAL TESTNET RUN · 25 OCT 2026 · 18:42 UTC	
Close transaction	0626430c…5ca1da ↗
Status · ledger	SUCCESS · 5386036
Payment channel	CBFRK6…XHKM ↗
Deposit locked by the fleet	1.00 USDC
Paid to the station (withdraw event)	0.30 USDC
Back to the fleet (refund event)	0.70 USDC
Independent auditor, V01–V08	PASS

8 SCENARIOS · ALL AS EXPECTED	
S1 · Happy path	PASS
F1 · Tampered meter reading	Refused
F2 · Fleet budget cap	Refused
F3 · Replay and duplicates	Refused
F4 · Payment detached from evidence	Refused
F5 · False or missing settlement	Auditor FAIL
F6 · Edited records	Auditor FAIL
F7 · Lost connection	Pending

```
$ stellar tx fetch result --hash 0626430c…5ca1da --network testnet → status SUCCESS, ledger 5386036
```

What we prove — and what closes each gap next.

WE PROVE

- Readings unchanged since the meter signed them
- The payer approved this exact payment
- The price replays exactly, to the base unit
- The money settled on Stellar Testnet

Not claimed yet	What closes the gap next
A certified meter — ours is simulated	Plug into certified meters that already sign their readings
Meters that sign only start and end readings	Provisional amounts reconciled at the end (roadmap)
An audited channel — it's experimental, Testnet only	Scout Soroban and Audit Bank before any mainnet use
An audit trail beyond RPC's ~7 days	Stellar indexers: Mercury, Goldsky, OBSRVR
Always-on channel watching	OpenZeppelin Monitor and Relayer
Private amounts — channels are public	Confidential settlement, as a separate research path
Many meter types — one adapter today (Wh)	A second, genuinely different meter next

Hard questions, short answers.

“Isn’t this just MPP?”

MPP moves the money; UsagePay proves what it was for. We use MPP unchanged and add the proof on top.

“Can you prove the energy was delivered?”

We prove the reading wasn’t changed after the meter signed it. Physical truth comes from certified meters — we plug into them.

“Why a blockchain, not invoices?”

Both companies — and any outsider — can verify the settlement themselves. We don’t claim cost savings yet: a pilot will measure them.

“How would you make money?”

Two separate hypotheses: an open-source evidence core, and managed channel operations for operators. No validated buyer yet — a pilot comes first.

“What about privacy?”

Stellar channels are public, so amounts are visible. The evidence stays off-chain; confidential settlement is separate research.

“Doesn’t the deposit lock capital?”

Yes. It’s a real cost on the payer’s side, and our economics model counts it instead of hiding it.